

Results

Results I

Results are interpreted only after the registered analysis has run and the deviation ledger ([Section @sec:deviations]) has classified any departures from the plan. Every value below is read from `output/data/demo_analysis.json`, which is regenerated by `scripts/generate_figures.py` from the tested `run_registered_analysis` function; the sensitivity row in the final subsection is read from the frozen registration's `sensitivity_analyses`, the same source `build_review_packet` consumes.

Confirmatory outcome (H1, primary_score) |

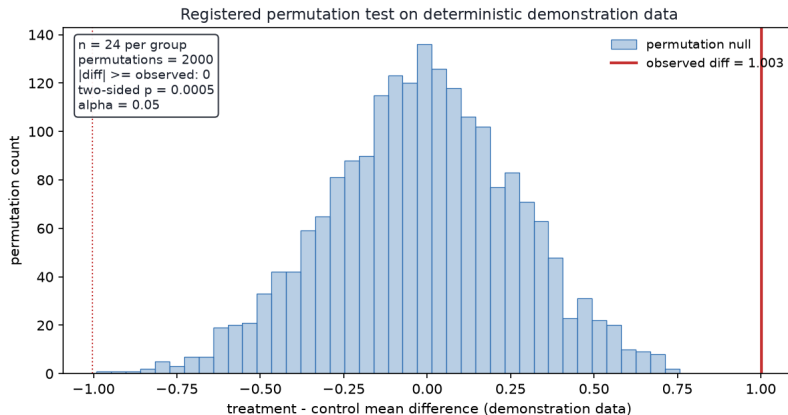
On the deterministic demonstration data, the registered two-sided permutation test returns:

Quantity	Value
Control mean	-0.178991
Treatment mean	0.823948
Observed mean difference	1.002940
Permutations	2000
Shuffles with absolute difference observed	0
Two-sided permutation p-value	0.0005
Preregistered alpha	0.05
Confirmatory decision	significant

Confirmatory outcome (H1, primary_score) ||

The observed difference of 1.003 lies far in the right tail of the permutation null distribution: none of the 2000 label shuffles produced a mean difference as extreme, so the add-one-corrected p-value is 0.0005, below the preregistered $\alpha = 0.05$. H1 is therefore **supported** as a confirmatory result. [fig:permutation_result] shows the observed statistic against the null it was tested against.

Confirmatory outcome (H1, primary_score) III



Confirmatory outcome (H1, primary_score) IV

Figure 1: Registered permutation test on the deterministic demonstration data, rendered by `plot_permutation_result` from `permutation_null_distribution` and `run_permutation_test`. The histogram is the label-permutation null distribution of the treatment-minus-control mean difference (2000 seeded shuffles); the solid red line marks the observed difference of 1.003 and the dotted line its mirror image. The annotation reports $n = 24$ per group, 0 shuffles at least as extreme, a two-sided p-value of 0.0005, and $\alpha = 0.05$. The observed statistic sits outside the entire null, which is why the primary confirmatory claim holds.

Claim boundary I

The registration preregisters exactly one confirmatory outcome. The review packet built by `build_review_packet` records `primary_score` as the sole confirmatory outcome and returns an integrity score of 1.0 for a plan-faithful execution. Any other endpoint examined during analysis is exploratory and is reported under [Section @sec:deviations], never here. This separation is enforced in code, not merely asserted in prose.

Sensitivity analysis I

The frozen registration declares one preregistered sensitivity analysis, and `validate_sensitivity_table` checks every declared row against the registered outcomes and primary model before the review packet is built:

Analysis	Target	Model	Decision
label_permutation_sensitivity	primary_score	permutation_test	robust

Sensitivity analysis II

The row targets the registered primary outcome with the registered primary model and a supported decision, so `validate_sensitivity_table` returns no findings for it and `output/reports/sensitivity_findings.json` records an empty findings list. The same row is carried verbatim into the review packet (`build_review_packet` reads `sensitivity_analyses` from the frozen plan), so the sensitivity check is visible in source, tests, review packets, and this manuscript — never invented after results are known.